

E-Commerce Transactions: Data Cleaning & EDA Insight Report

Online Retail Transactions | 1 Dec 2010 – 9 Dec 2011 | Python, Pandas, NumPy, Matplotlib, Seaborn

1. Business Objective

Clean and explore an e-commerce transaction dataset to identify patterns in sales performance, products, customers, markets and transaction trends, while validating data-quality issues before drawing business conclusions.

2. Data Preparation & Cleaning

The original dataset contained 541,909 rows. The analysis removed 5,268 exact duplicates, converted InvoiceDate to datetime, standardised selected fields, investigated missing values and anomalies, and separated cancellations, accounting adjustments and other non-standard records from standard sales analysis. A sales_value field was calculated as quantity × unit price. The resulting standard-sales dataset contained 524,877 transaction lines.

3. Key Performance Indicators

Metric	Result
Standard sales transaction lines	524,877
Total sales value	10,631,048.74
Unique invoices	19,959
Identified customers	4,338
Average transaction line value	20.25
Top 10 customer sales share	17.30%

Note: transaction lines are individual product lines; one invoice can contain multiple product lines.

4. Key Business Insights

1. Sales growth: November 2011 recorded the highest complete-month sales value at approximately 1.50 million and the highest transaction-line volume at 82,004. The increase appears primarily volume driven.
2. Market concentration: The United Kingdom generated approximately 8.73 million in product sales, substantially exceeding other markets. The Netherlands, EIRE, Germany and France were the next largest markets.
3. Customer concentration: The top 10 identified customers contributed 17.30% of sales attributed to customers with an identified CustomerID, indicating a meaningful contribution from high-value customers.
4. Product performance: REGENCY CAKESTAND 3 TIER generated approximately 174,156.54 in sales. PAPER CRAFT, LITTLE BIRDIE ranked highly but was heavily influenced by one 80,995-unit transaction and was therefore treated as an extreme outlier.
5. Data quality: Missing CustomerIDs, cancellations, accounting adjustments, zero-priced records and extreme quantities required business-aware treatment rather than automatic deletion.

5. Data Limitations

Approximately 25.16% of records have a missing CustomerID, limiting customer-level analysis. December 2011 is a partial month (data through 9 December). Extreme transactions may influence averages and rankings. The dataset also lacks demographic, marketing-channel and acquisition-cost information.

6. Business Implications

The business could investigate the drivers of the September–November sales increase, monitor the UK market while developing higher-value international markets, strengthen retention of high-value customers, review high-performing products for inventory and promotional planning, and improve CustomerID capture.

7. Conclusion

The analysis shows how disciplined data cleaning and exploratory analysis can reveal actionable patterns in e-commerce performance. The cleaned dataset provides a stronger foundation for further customer, product and sales analysis while making key data limitations explicit.